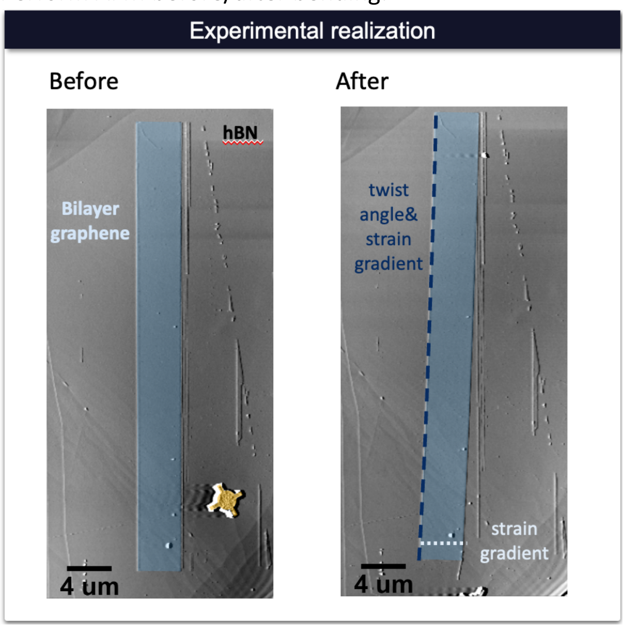
Outline& General idea

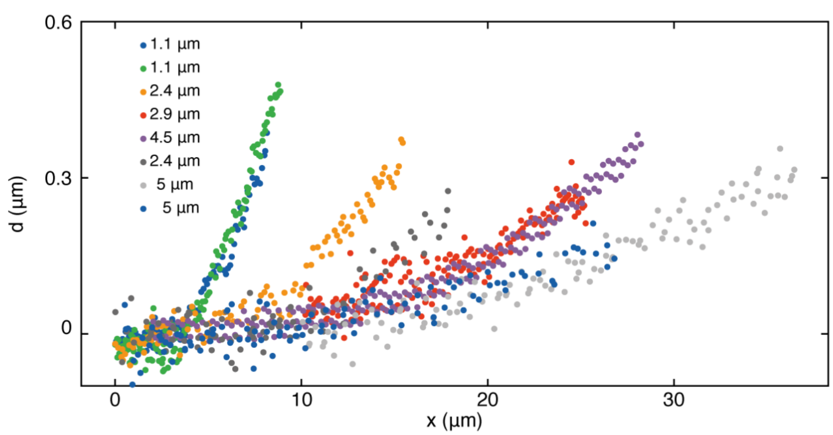
Thursday, June 4, 2026 10:44 AM

Overview

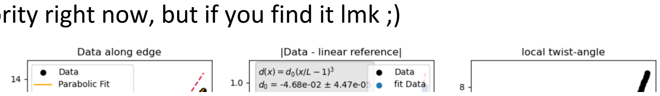
- Use AFM-tip to bend ribbons: https://www.science.org/doi/full/10.1126/science.ade9995
- Here, we use bernal bilayer graphene instead of monolayer graphene
- Perform AFM before/after bending:
- 

Open question : What is the bending statistics of bernal bilayer graphene?

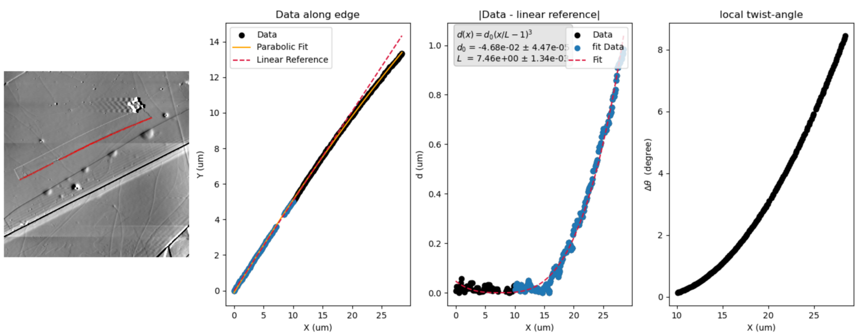
To answer this: Analyse the bend in the ribbon. Ultimately we would like to be able to plot a similar statistics plot for BLG as done in the supplementary of the paper for MLG



Practical how to: Analyse the bend along the edge before and after bending by using the analysis scripts that I am attaching. Analysis.py opens a GUI that will let you select points along the edges of the ribbons, and save them into a .txt. You can then work with this data (in example how to can be found in Analysis_Script_fit_v1.py), fit it , and try to extract displacement d and local twist angle. CAVEAT: there is currently a bug in the script wrt the twist angle analysis. Its not highest priority right now, but if you find it lmk ;)



not highest priority right now, but if you find it lmk ;)



Additionally: To improve this analysis, one could take the scan before bending & after bending in and calculate d from data_postBend - data_preBend instead of the way its currently done (|data-linearREf|). That is currently not implemented yet, but could be valuable. Caveat: some ribbons don't just bend but also rotate, for these ribbons this better analysis will not work.

DataFormat: .ibw - can be opened with Gwyddion (an open source AFM analysis platform that I would heavily recommend for quick and dirty checks of the AFM scan). I am also sending you the library script to load it into python (Asylum_loadData_v3.py).

DataDetails: I am sending you multiple samples, labelled with variations of ….cgXX where XX is a number, and a pre-bend and post-bend scan for each. I was not super consistent with labelling, but lower number or sth with "before"/ "pre" is scan before bending, "after" or later number is post bending

Proposed steps
(1) Once you are familiar with Gwyddion and the python scripts, I would start by analysing the ribbon edge bending as done in the Analysis scripts I send you & plot the results into a similar figure as the one from the supplementary of the bending paper attached above. To do so, find the edge of the ribbon in the AFM scans, and determine which & how many points are along the straight edge of the ribbon and should thus be included in the linear fit that serves as reference for the d(x) analysis. Caveat: this straight edge can be on either side of the ribbon. You usually can see which side I bent on through the presence of this little gear-shaped rotator. The straight edge should be the edge at the further end of the ribbon.
(2) once this is done, there is a couple of possible next steps:
- Optimize the code by getting d(x) through the different before and after bending instead of the (somewhat judgement-prone) linear-reference
- Fix the twist angle analysis

(3) Ultimately we would like to compare this to the MLG case & find similarities and differences. I think this should be done throughout, once you have done (1) and see if we can improve it via(2).

Please let me know if you have any questions at any time or if something is not clear. Thank you so much for your help with this.